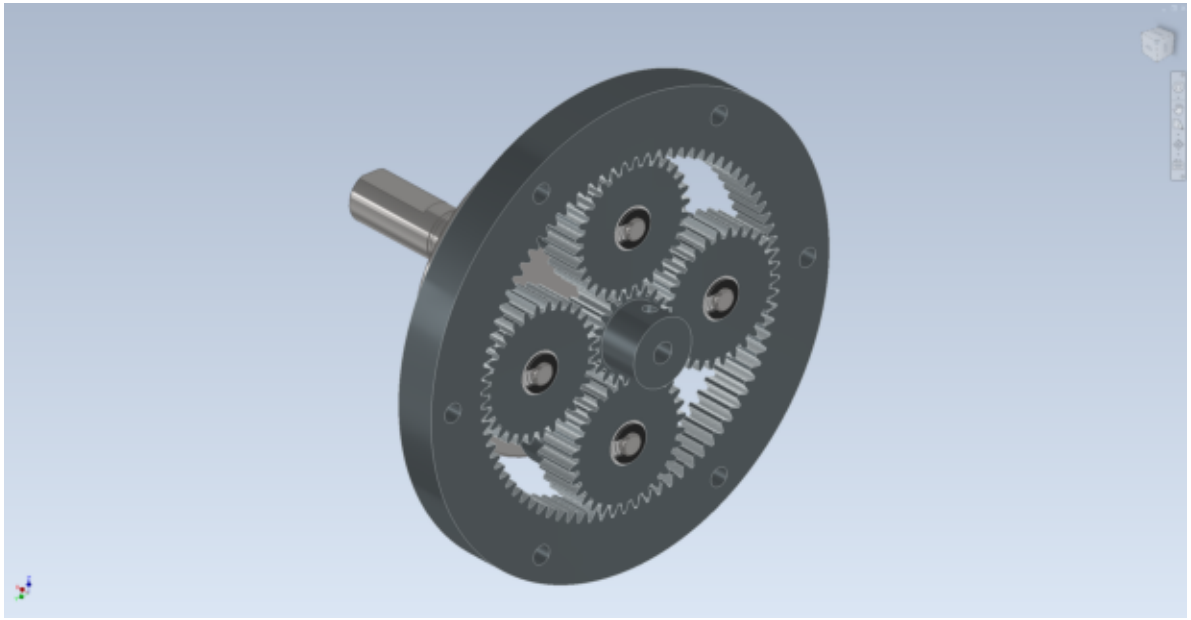
Planetary Gearbox 5:1 for a NEMA 17 stepper

PlanetaryGearbox - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

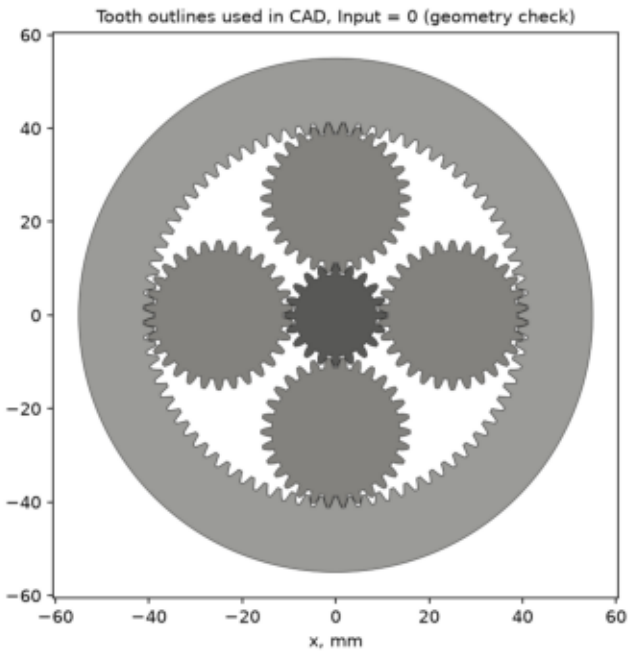


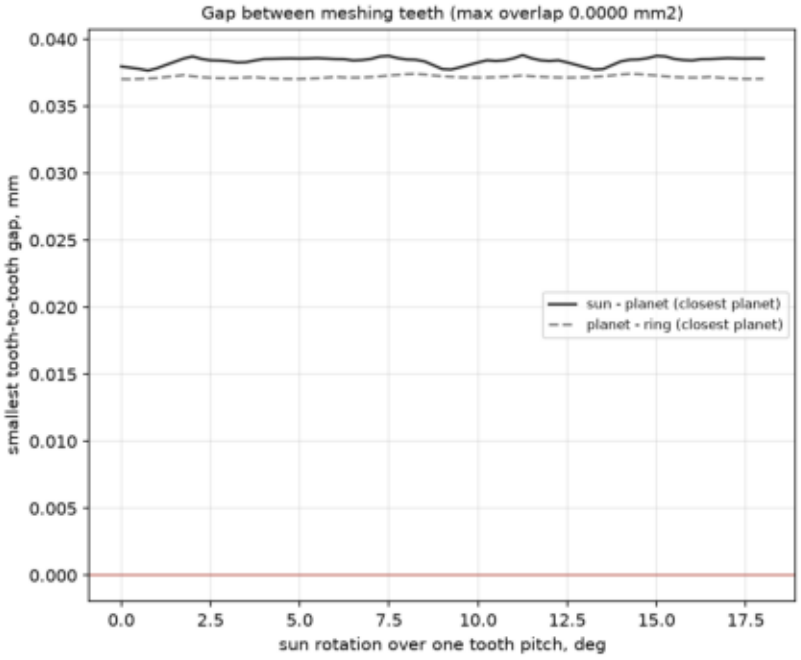
Item	Value	Source
Gear train	KHK m1, 20 deg, S45C: sun SS1-20A (z20), 4 x planet SS1-30 (z30), ring SI1-80 (z80) fixed	CAT
Ratio	1 + 80/20 = 5.0, centre distance 25 mm	CALC
Motor	StepperOnline 17HS19-2004S1, 0.59 Nm holding, 5 mm D-shaft	CAT
Rated output torque	5.13 Nm continuous (limit: KHK surface sun)	CALC
Motor-limited torque	2.83 Nm at holding torque; 1.89 Nm at 60 rpm out	CALC
Efficiency	94.6 % at the design point, 98.8 % tooth friction only	CALC
Backlash at output	12-28 arcmin (KHK tolerances), 35 worst case	CALC
Mass	2.17 kg with motor, 1.78 kg gearbox, bracket 0.25 kg	CAD
Assembly	48 occurrences, 141 constraints, 0 DOF; 0 clashes at 9 poses	CAD
Envelope	143 x 132 x 128 mm incl. bracket	CAD

1. Tooth counts, geometry and kinematics

PlanetaryGearbox - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Check	Result	Source
Ratio (sun in, ring fixed, carrier out)	1 + Zr/Zs = 1 + 80/20 = 5.00	CALC
Concentricity Zr = Zs + 2 Zp	80 = 20 + 2 x 30, so a = m (Zs + Zp)/2 = m (Zr - Zp)/2 = 25 mm	CALC
Assembly condition (Zs + Zr)/N	(20 + 80)/4 = 25, an integer: 4 equally spaced planets fit	CALC
Neighbour clearance, 4 planets	2 a sin(180/4) - da = 3.36 mm between planet tips	CALC
Same check with 5 planets	-2.61 mm: tips overlap, so 5 is impossible (and 100/5 = 20 would allow it)	CALC
Contact ratio sun-planet / planet-ring	1.61 / 1.94	CALC
Speeds at 300 rpm motor	carrier 60 rpm, planet -100 rpm absolute, 160 rpm on its pin	CALC
Pitch-line speed (relative to carrier)	0.251 m/s	CALC
Tooth gap in the 2D outline check	0.0377 mm sun-planet, 0.0370 planet-ring, overlap 0.0 mm2 over 80 poses	CALC
Tooth gap measured in Inventor	0.038 mm sun-planet, 0.037 mm planet-ring (min over 9 poses)	CAD
Kinematics in the assembly	carrier = Input/5 and planets = -Input/3 checked at 9 poses: max error 0.0 deg	CAD



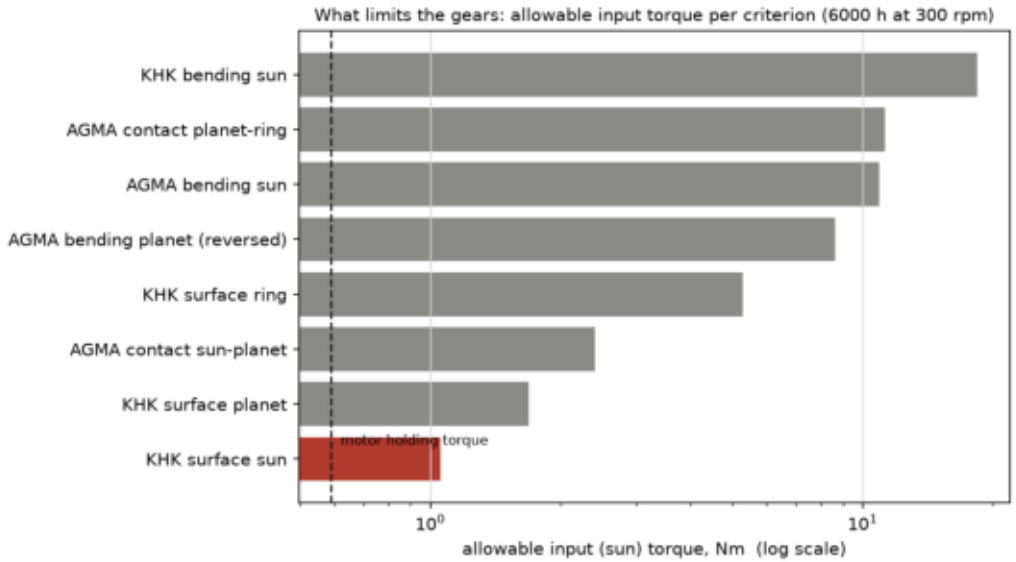


2. Gear strength: Lewis, AGMA 2001 and the KHK catalogue

PlanetaryGearbox - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

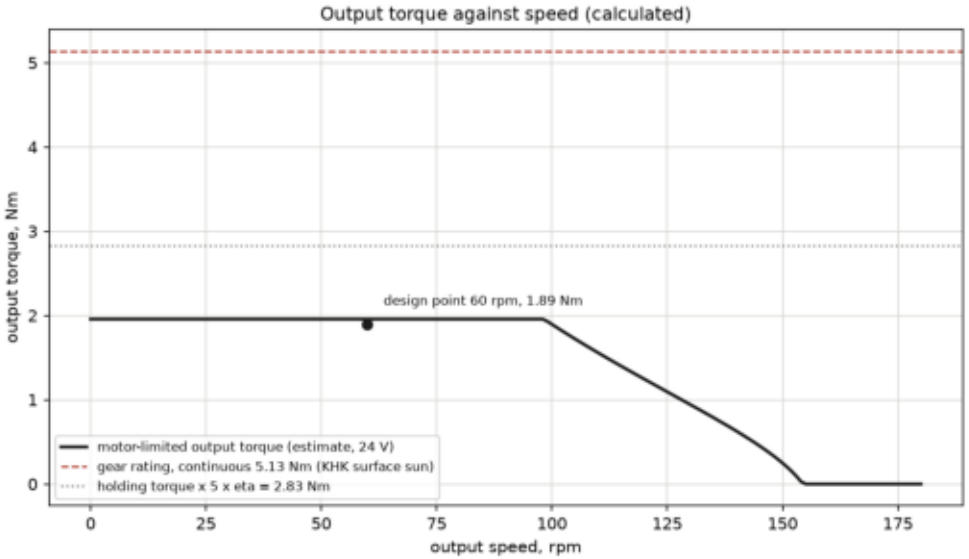
Quantity	Design 0.40 Nm in	Holding 0.59 Nm in	Source
Tangential force per mesh (Kgamma 1.25)	12.5 N	18.4 N	CALC
Lewis bending sun / planet / ring	4.0 / 3.6 / 2.7 MPa	6.0 / 5.3 / 4.0 MPa	CALC
AGMA bending sun / planet	6.4 / 5.9 MPa	9.5 / 8.7 MPa	CALC
AGMA bending SF sun / planet (x0.7 reversed)	27.3 / 21.5	18.5 / 14.6	CALC
AGMA contact sun-planet / planet-ring	201 / 100 MPa	244 / 122 MPa	CALC
AGMA contact SH (on stress)	2.45 / 5.30	2.02 / 4.36	CALC
KHK surface durability SF (torque) sun / planet / ring	2.64 / 4.21 / 13.18	1.79 / 2.86 / 8.94	CAT/CALC
KHK bending strength SF (torque) sun	46	31	CAT/CALC

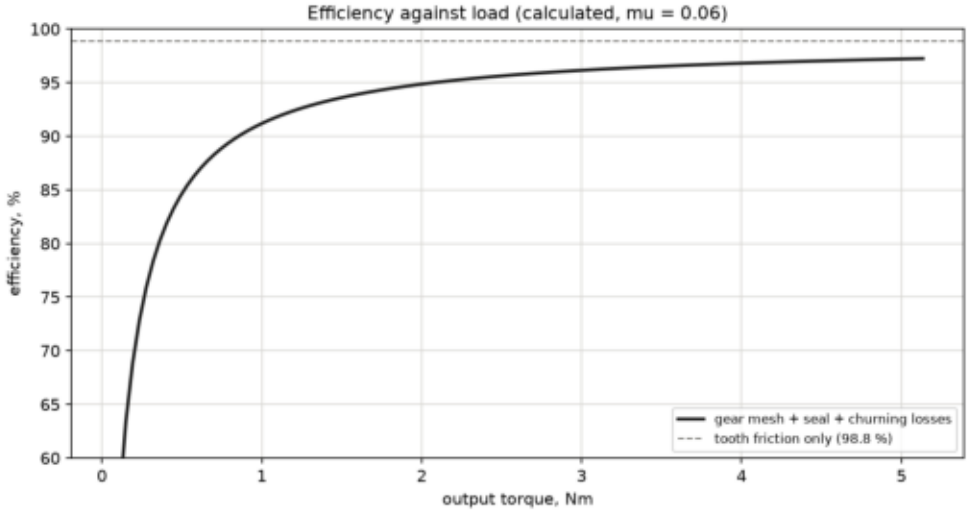
Inputs: S45C assumed 180 HB (KHK: below 194 HB); AGMA grade 1 through-hardened St = 184 MPa, Sc = 600 MPa; Ko 1.25, KH 1.28, Kv 1.062 (Qv 8), J 0.33 / 0.36, ZE 191; life 6000 h at 300 rpm (the motor's rated TMBF) gives 3.46e+08 sun cycles, ZN 0.82, YN 0.96. The planet is an idler, so its teeth bend both ways: allowable x 0.7. Decision: the AGMA contact check allows 2.40 Nm at the sun, but KHK's own catalogue lists the SS1-20A at 0.33 Nm surface durability, which allows 1.06 Nm with 4 planets and Kgamma 1.25. KHK does not print its calculation conditions on the product page, so I rated the gearbox on the lower catalogue number: 5.13 Nm continuous at the output. The motor can only deliver 2.83 Nm through it at holding torque, a 1.79x margin on the rating. Bending is never close: SF above 14 everywhere.



3. Torque, speed and efficiency

Motor model (estimate, not the StepperOnline pull-out curve): Kt = 0.59 Nm / 2 A = 0.295 Nm/A, Ke taken equal to Kt, 24 V chopper, R 1.6 ohm, L 3.0 mH, 50 pole pairs, running torque = 0.7 x holding. Gives 0.41 Nm at 300 rpm, 0.27 Nm at 600 rpm, zero near 780 rpm. Efficiency: Niemann mesh-loss factors HV 0.179 (sun-planet) and 0.064 (planet-ring) with mu 0.06 give 98.93 % and 99.61 % per mesh, 98.54 % for the fixed-carrier train and (1 + 4 x eta0)/5 = 98.84 % for the reducer. Seal drag of the two 6001-2RSH (0.059 Nm, SKF model class values) and 0.005 Nm churning at the input pull it down to 94.6 % at the design point. Heat: 0.67 W lost at 12.6 W in, nothing to cool.





4. Bearings, planet pins, output shaft and joints

PlanetaryGearbox - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Item	Result	Source
Planet bearings NSK MR105ZZ (C 475 N, C0 168 N)	pin load 36.9 N at holding, inner bearing 30.0 N: static SF 5.6, L10 415091 h at 160 rpm; at the rating 54 N, SF 3.1	CALC
Output bearings SKF 6001-2RSH (C 5.4 kN)	overhung load 200 N at 17 mm: outer bearing 412 N, L10 623171 h at 60 rpm, static SF 5.7; 940 N would still give 6000 h	CALC
Planet pin 5 mm hardened, cantilever	83 MPa at the rating, SF 17 on 1400 MPa yield (assumed); deflection 13 um at the gear	CALC
Output shaft 12 mm 1045, groove + flat	rating torque + 200 N: 20 MPa shear at the flat, fatigue SF 3.5 at the circlip groove (Kf 2.8), yield SF 16 at the bearing seat	CALC
Ring clamp, 6 x M4 12.9	ring reaction 4.2 Nm at the rating vs 353 Nm friction (4 kN preload, mu 0.15, 2 faces): SF 84; thread engagement 10 mm = 2.5 d in 6061	CALC
Sun on the motor shaft	engagement 19 mm on the 15 mm D-flat; sun centred 20 mm from the flange, where the datasheet allows 21 N radial; planet load sharing error gives 7.8 N (SF 2.7)	CALC/CAT
Static peak output	16.5 Nm loads the inner MR105 to its C0 (the weakest static item); the sun teeth would take 92 Nm	CALC

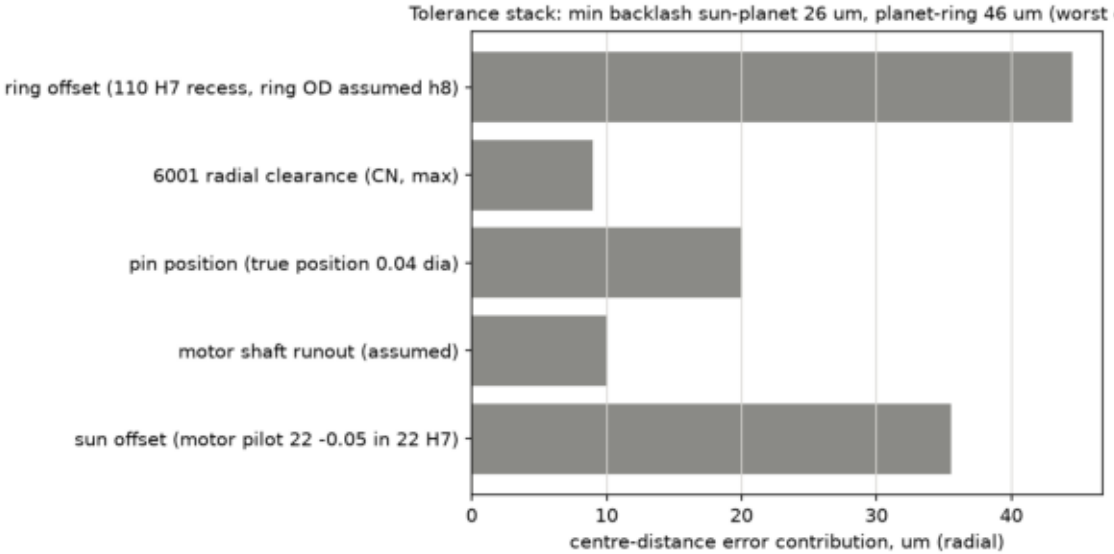
Decision: KHK's SS1-30 has a plain 10 mm bore, which takes an MR105ZZ (5 x 10 x 4) directly. Two per planet with Loctite 641 means the planets are used exactly as bought; the only change to any catalogue gear is six drilled holes in the ring. The MR105 static capacity then sets the short-term peak, not the gear teeth.
Decision: with 4 planets there is only 3.1 mm of radius between neighbouring planet tips at the gear face, too little for a second carrier plate's pillars, so the pins are cantilevered from one 1045 carrier plate. The pin bends 13 um at the rating, which is small against the 80-180 um backlash.

5. Backlash and tolerance stack

PlanetaryGearbox - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Contribution (radial)	Value (CALC)
sun offset (motor pilot 22 -0.05 in 22 H7)	35.5 um
motor shaft runout (assumed)	10.0 um
pin position (true position 0.04 dia)	20.0 um
6001 radial clearance (CN, max)	9.0 um
ring offset (110 H7 recess, ring OD assumed h8)	44.5 um
Worst-case centre-distance loss, sun-planet / planet-ring	75 / 73 um
RSS, sun-planet / planet-ring	43 / 50 um
Min backlash left (KHK min - 2 da tan20), sun-planet	26 um (first layout with a 22.05 pilot: 4 um)
Min backlash left, planet-ring	46 um
Output lost motion (j1 + j2)/(2a)	12.4-27.5 arcmin, 34.9 worst case

Decision: the first adapter had a 22.05 mm clearance recess for the motor's 22 -0.05 pilot. The stack showed that this play, plus pin position and bearing clearance, could close the sun-planet backlash to 4 um, close to binding. The pilot is now 22 H7 (+0.021), which leaves 26 um worst case. The ring is centred by 110 H7 recesses in both housing halves rather than by its bolts.

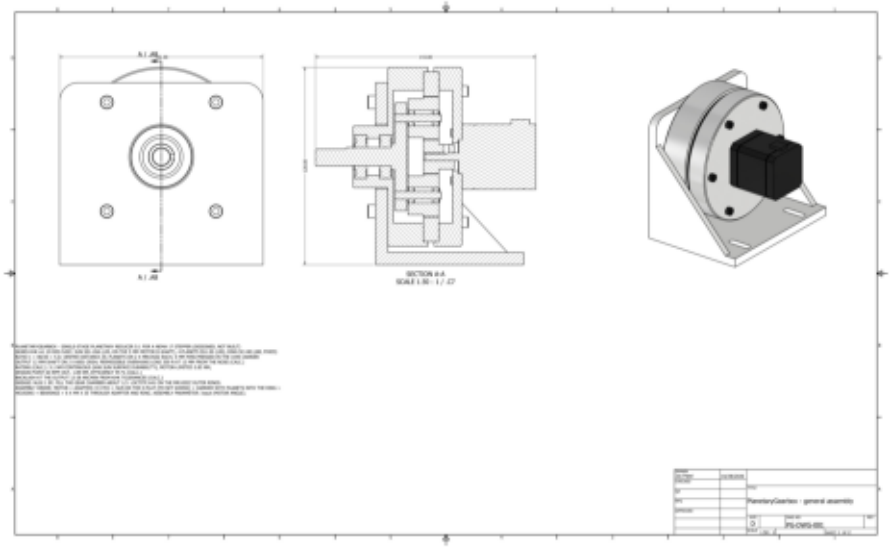


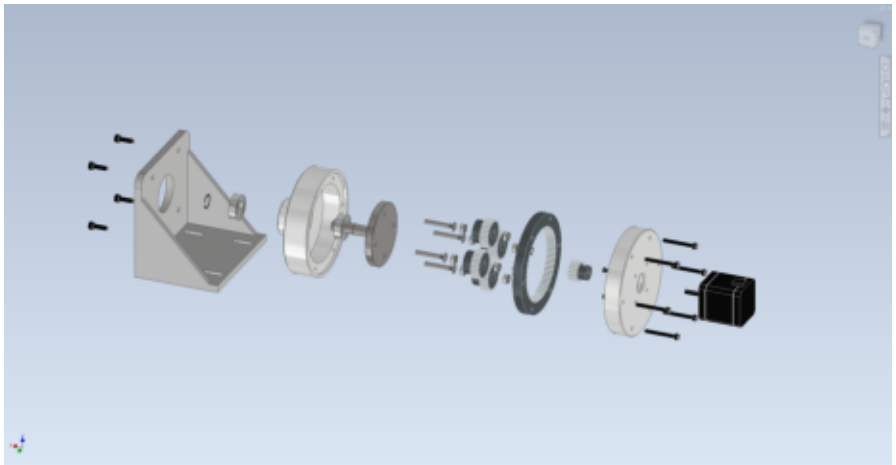
6. Inventor checks, printed bracket and mass

PlanetaryGearbox - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / datasheet.

Check	Result	Source
DOF / constraints	0 DOF, 141 constraints {'Flush': 117, 'Mate': 18, 'Angle': 6}, 0 unhealthy	CAD
Interference	0 clashing pairs at Input = 0, 4.5, 9, 13.5, 27, 90, 133, 247, 341 deg	CAD
Planet to housing / adapter	2.03 / 2.03 mm (iteration 1 had 0.00 mm, see the decision below)	CAD
Planet to planet, planet to carrier	3.37 mm, 0.50 mm (shim gap)	CAD
Gear masses vs KHK catalogue	PG-211 36.2 g vs 37 g; PG-212 80.7 g vs 82 g; PG-213 342.4 g vs 350 g	CAD / CAT
Bracket (PETG), printability	watertight, min wall 4.2 mm, support 0.2 % of the area, fits the 256 mm bed, best of 6 orientations	CALC (dfam)
Bracket, strength	2.17 kg at 29 mm: 0.44 MPa (SF 45 on 20 MPa printed); hole bearing 1.3 MPa at the rating torque (SF 16)	CALC
Mass	gears 0.702, carrier group 0.304, housings 0.676, motor 0.390, bearings 0.054, fasteners 0.047, bracket 0.252 kg	CAD

Decision (CAD clearance check, iteration 1 -> 2): with 80 mm bores in the adapter and housing, the planet tips (orbit radius 41 mm) ran on the faces that seat the ring: 0.00 mm clearance, metal rubbing on every turn. Both bores went to 86 mm (2.0 mm radial clearance) and the M5 mounting pattern moved from 88 to 100 mm PCD to keep 4.5 mm of wall around the bigger bore. All checks were re-run; iteration 1 results are kept in Tests\Assembly_Checks\Iteration1.





7. Assumptions, sources and limits

Catalogue data (checked online 2026-10-28): KHK SS1-20A (bore 5, hub 16, 20 long, bending 5.75 Nm, surface 0.33 Nm, 0.037 kg), SS1-30 (bore 10, hub 25, 10.1 / 0.79 Nm, 0.082 kg), SI1-80 (110 x 78 x 10, 38.8 / 6.59 Nm, 0.35 kg, $91.64 list) from catalog.khkgears.us; StepperOnline 17HS19-2004S1 drawing A0842 (0.59 Nm, 2.0 A, 1.6 ohm, 3.0 mH, 82 g cm2, shaft 5 x 24 with 15 mm D-cut, 22 pilot, M3 x 4.5 on 31, 21 N radial at 20 mm, TMBF 6000 h at 24 V / 300 rpm); NSK MR105ZZ (475 / 168 N, nsk.com); SKF 6001-2RSH (5.4 / 2.36 kN, distributor listings); DIN 471 12 x 1 (McMaster 98541A119).
Assumed: gear hardness 180 HB; AGMA factors Ko 1.25, Kgamma 1.25 for 4 planets with no floating member, J 0.33 / 0.36; tooth friction 0.06; seal drag from SKF model class values; motor torque-speed from an R-L-back-EMF model with a 0.7 running factor (not the manufacturer's curve); motor mass 0.39 kg from reseller listings; ring OD tolerance h8 (KHK does not list it); pin yield 1400 MPa; PETG printed allowable 20 MPa.
Not checked: KHK's own calculation conditions for its strength values (not on the product pages); thermal growth; grease life; noise; the stepper driver and power supply (listed in the BOM, not modelled). The motor mass and connector shape are envelopes.
What could go wrong first when built: the sun-planet contact (unhardened S45C) if the gearbox is run near its rating for long, and the MR105 bearings under shock loads above about 16 Nm at the output. A hardened sun (KHK SSA / SSG class) would be the first upgrade if more torque is needed.
Files: Analysis\gearbox_analysis.py (all calculations), Tests\Gear_Mesh\mesh2d.py (tooth outline check), _build*.py (Inventor model, checks, drawings, renders), Tests\Buildability\buildability_check.md (parts, fits, assembly order, cost).